2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS **[2 Hours]**

SCC.212 Advanced Programming

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 20 marks.

Use a separate answer book for each question.

Question 1

Imagine you have been asked by a large government organization to write an extensible Java program capable of modelling a democratic national election. In such an election, a number of voters may choose to vote for one of a fixed number of candidates. At the end of the election, the candidate with the most votes is declared the winner.
You have been given the following set of requirements:

- The choice of candidates should be easily definable within the program in a class called **Election**. Candidates are identified by a plain text name.
- A **Vote** class should record the candidate chosen by a given **Voter**.
- A **Voter** is identified by a unique number. A **Voter** may vote, strictly once, through an explicit **vote**() method in the **Voter** class. Attempting to vote multiple times in any given **Election** should result in a custom **Exception** being thrown. Any attempt to vote for a candidate with an unrecognised name should be recorded as a spoiled vote.
- The **Election** class stores all the votes cast during the election. The number of votes that may be cast is potentially very large so an appropriate class from the Java Collections API should be used. The **Election** class should also calculate the final results, including a printout to the screen of the number of votes each candidate was awarded, along with the number of spoiled votes.

1.a Write a Java program to meet the above requirements, and also write a simple test program in a main() method that illustrates how your program would operate. Minor errors in syntax will not be penalised, but **pay particular attention to good OO design principles throughout**.

[12 Marks]

1.b Your client has asked that you also now support proxy voting, such that **Voter A** may also **vote** on behalf of **Voter B,** but only if **Voter B** has explicitly assigned **Voter A** as a proxy. Any attempt to illegally vote on behalf of another voter should result in a custom **Exception** being thrown. Show how you would adapt your program with **minimal changes** to meet this new requirement.

[4 Marks]

1.c Your client appears with yet another requirement. They wish to also record postal voting. A **PostalVote** should act exactly as an existing **Vote**, but also record the date at which the postal vote was made. Show how you would adapt your program with **minimal changes** to meet this new requirement.

[4 Marks]

Total 20 marks

Question 2

2.a Programming languages contain a range of concepts that allow programmers to write more elegant, scalable and extensible programs. For each of the following concepts, provide a concise description of what it is, and where applicable, describe the mechanisms that enable its use in both Java **and** Erlang.

i. Immutability
ii. Method overloading
iii. Polymorphism
iv. Encapsulation

[12 marks]

2.b Write an Erlang function **from first principles** (using no built in functions) that takes two parameters: a list of items and a value. The function should implement a basic search algorithm and return the number of times the given value is found in the list.

[4 marks]

2.c State if your program is head recursive, middle recursive or tail recursive. If your program is **NOT** already tail recursive, rewrite it such that it is tail recursive.

[4 marks]

Total 20 marks

Question 3

3.a *"Python variables do not have to be explicitly declared to reserve memory space".* Briefly explain how Python variables are created.

[2 marks]

3.b What is the final result of executing the following Python programs?

i.

```
s = lambda p,q: lambda x,y: (p-q)*(x+y)
g = s(2,3)
print(g(4,6))
```

ii.

```
s = 'Lake District'
p = range(1,len(s)-1, 3)
p = list(p)
print(p)
```

iii.

```
s = 'Lancaster University'
s = s[-24:-16]+s[8:]
print(s)
```

[6 marks]

3.c Consider the following matrix, **mat**:

```
        #  c1 c2 c3 c4 c5 c6
mat = [[ 2, 4, 7, 5, 3, 8 ],
       [ 6, 1, 9, 0, 2, 5 ],
       [ 3, 0, 5, 1, 8, 2 ],
       [ 4, 7, 2, 6, 9, 0 ],
       [ 9, 5, 3, 7, 8, 4 ],
       [ 8, 2, 1, 4, 1, 6 ]]
```

Write a Python program that uses *list comprehension* and the *reduce* function to generate and print the product of even numbered columns in **mat** (i.e. *c2, c4* and *c6*). Column tallies (totals) should exclude all integers greater than '**7**'.

[6 marks]

Question 3 continues on next page...

Question 3 continued...

3.d The *compute* function shown below takes two numbers and an arithmetic operator, and returns the result of performing the computation. Rewrite the function logic to achieve the same result without using **if**, **elif**, **else**, additional **functions** or **class** statements.

[6 marks]

```
import random

def compute(a,operator,b):
    if (operator == '+'):
        result = a + b
    elif (operator == '-'):
        result = a - b
    elif (operator == '/'):
        result = a / b
    elif (operator == '*'):
        result = a * b
    return result

#main program
for i in range(5):
   a = random.randint(5,9)
   b = random.randint(1,5)
   operator= random.choice('+-*/')
   result = compute(a,operator,b)
   print(a,operator,b,'=',result)
```

Total 20 marks

Question 4

4.a Write a Python class that takes as input, *day of the month*, *month* and *year*, and creates a **Calendar** object.

Your program should ensure the following requirements:

i. *Day of the month*, *month* and *year* must be integers.
ii. *Year* must be 4 digits.
iii. The *day of the month* must correct for the month (e.g. 30/**2**/2017 is incorrect)
iv. *Month* must be correct for the year (e.g. 29/**2**/2017 is incorrect)
v. Provide a method for returning the date string in the form '*day/month/year*'.
vi. Provide a method for advancing the date that takes into account leap years. When the advance method is called on a date object, it should return the next day's date.

The number of days in each month for the year 2017 is provided in the table below.

Jan	Feb	March	April	May	June	July	Aug	Sept	Oct	Nov	Dec
31	28	31	30	31	30	31	31	30	31	30	31

[16 marks]

4.b Rewrite the following the *Java* methods as *Python* **lambda** functions

i.
```
public boolean methodA(int x, y)
{
    return (x%6!=1 && y>4);
}
```
[2 marks]

ii.
```
public boolean methodA(int x, int y)
{
        methodB(x+y);
}
```
[2 marks]

Total 20 marks

--- End of Paper ---